

Key Feature	Use
KPI, Report, and Stories Visualization	With this, you can: • Visualize and comprehend data from different virtual data models that represent different business areas. • Configure business metrics, interpret and interact with your data in real-time; visualize, and analyze the data. • Create reports for the same KPIs. • Integrate and visualize embedded stories for data analysis

2.14 Enterprise Technology

When SAP S/4HANA applications are deployed on an SAP HANA runtime database, all Enterprise Technology solutions made available to you under your SAP S/4HANA application license/subscription are “Runtime Software” (i.e. as defined in the applicable SAP Software Use Rights document), and thus may only be used to enable performance of the licensed/subscribed to SAP S/4HANA applications. As such, and for the avoidance of doubt, application(s) and/or technology(ies) that are not licensed/subscribed to from SAP may NOT access the SAP HANA runtime database via any Enterprise Technology solution(s); rather, such access is only permissible via the SAP S/4HANA application layer.

2.14.1 ABAP Platform

ABAP platform is the foundation for ABAP-based SAP solutions that build on top of the database and data management platform of SAP HANA.

2.14.1.1 Operating the Application Server for ABAP

Administrators can use the concepts and functions of the SAP Application Server for ABAP (AS ABAP) as a basis for the monitoring and administration of AS ABAP.

Key Features

The following features are available:

General Administration Tasks

Key Feature	Use
Managing SAP licenses	Assign license keys for SAP systems, determine the number of users and used engines of the SAP software for each system, and consolidate measurement results of one or more systems.
Configuring AS ABAP	Configure AS ABAP using profile parameters. For central scenarios you do not need to explicitly set parameter values, since defaults are set and values adapt automatically to physical conditions. Manual configuration of additional scenarios is possible.
Starting and stopping AS ABAP	Start and stop individual instances of your AS ABAP system or the complete system.
Managing and monitoring your AS ABAP within your system landscape	Use SAP Cloud ALM and SAP Focused RUN to manage and monitor several AS ABAP systems centrally within your system landscape. The functional scope of SAP Cloud ALM and SAP Focused RUN that is supported for SAP S/4HANA depends on your SAP maintenance contract: See also [https://support.sap.com/en/alm/usage-rights.html].
Providing and managing hosts, instances, databases, or systems	Provide, monitor, and control any host, instance, database, or SAP system centrally.
Managing network access to your SAP systems and configuring load balancing	Configure the access you want to allow to your SAP network and define access points for Web requests to your network. Such requests can be distributed (using load balancing) between individual Application Servers of your SAP system.
Printing	SAP provides its own spool service and a spool database for printing.
E-mailing	Send e-mails from and receive e-mails within the AS ABAP. Streamline your e-mail correspondence by creating custom templates based on predelivered templates.
Scheduling batch jobs	Schedule local batch jobs in your AS ABAP system.
Backing up and restoring your SAP system	Back up and restore your SAP system on operating system level and database level, if required.

Key Feature	Use
Archiving your data	Use the data archiving functions to archive any completed business transactions that are no longer relevant for your daily operations, and so significantly reduce the load on the database. SAP S/4HANA uses the functions for archiving provided by the ABAP platform.
Enabling your system for SAP remote support	Set up a remote service connection for each ABAP system to use SAP remote services (for example, SAP EarlyWatch or Remote Consulting), or if you would like to permit an SAP support consultant to work directly in your system to make a more precise problem diagnosis.

Local Administration Tasks for AS ABAP

Key Feature	Use
Exchanging the kernel without system downtime	Use an automated procedure to automatically exchange the kernel in an ABAP system, or to make parameter changes while the system is running.
Monitoring components of AS ABAP	Check the consistency of an installation, get an overview of all application server instances and their states, monitor processes, users, lock entries, Internet requests, RFC connections, and RFC calls, and cross-component processes.
Configuring and administering special AS ABAP components	Configure asynchronous RFCs, logon load balancing, operation modes, area menus, and the number range buffer. You can also send important notifications to all users that are logged on to the ABAP system in the form of short messages. It depends on the UI technologies whether these messages are consumed.
Troubleshooting	Use the trace function, the ABAP dump analysis, and the system log to identify and solve issues.
Managing concurrent database access	The SAP system is equipped with a special lock mechanism that synchronizes database access. The purpose of the lock mechanism is to prevent two transactions from changing the same data in the database simultaneously. You can configure, administer, and monitor database updates related to ABAP transactions.

Key Feature	Use
Managing the available memory of your ABAP system	Use profile parameters to configure SAP Memory Management in order to adjust the ABAP Server to hardware and software requirements.
Managing background processing	Configure and monitor background processing of the AS ABAP, and schedule and manage local background jobs.

Other Administration-Related Services for AS ABAP

Key Feature	Use
Managing alerts	Inform responsible or interested parties immediately via a predefined alert if the application detects a critical situation and triggers the alert management to create the alert.

Database Administration for SAP HANA

Key Feature	Use
Starting and stopping the SAP HANA database	Start and stop the SAP HANA database.
Monitoring and administering the SAP HANA database	Monitor and administer local or remote SAP HANA databases in SAP systems in an ABAP environment.

Software Logistics

Key Feature	Use
Managing and transporting development objects	Organize ABAP development projects and transport changes between SAP systems in your system landscape. In addition to the classic Change and Transport System (CTS), you can leverage Git repositories to store ABAP content and to provide it to ABAP systems. You can also copy ABAP clients and transport languages. Similar to ABAP objects, you can also transport SAP-specific non-ABAP applications in your system landscape. A close integration for transporting development objects of ABAP and SAP HANA is provided.

Other Tasks

Key Feature	Use
Maintaining Employees	Maintain business users with business role "Employee", with a workplace address, and a user assignment.

2.14.1.2 ABAP Large Language Model Access

Business Background

Artificial intelligence has become a powerful tool in today's economy. This is also true for software development. Thus, the ABAP Platform provides access to a variety of Large Language Models (LLMs) to increase efficiency and productivity of modern software development.

Key Features

The following features are available:

Key Feature	Use
Accessing LLMs	Accessing a variety of different LLMs to build AI-based applications.
Communicating with LLMs	Sending prompts to an LLM and receiving its response.
Configuring LLMs by setting their parameters	When supported by the LLM you use, you can set its parameters to leverage the LLM the best way possible to fit for your application.
Using Media Input	When supported by the LLM you use, you can add media input (images, documents, etc.) to your LLM requests.

2.14.1.3 Custom Application Development for ABAP

Business Background

A key ingredient for business applications is custom development. The ABAP development environment offers a wide range of options for custom application development from extensions and modification of SAP standard applications by customers up to the development of completely new applications.

Key Features

The following key features are available:

ABAP Language

ABAP is a fourth-generation (4GL) as well as an object-oriented programming language developed and constantly improved/enhanced by SAP for programming business applications in the SAP environment.

Key Feature	Use
Internal tables	Store and process mass table data dynamically in the working memory.

Key Feature	Use
Managing updates of data in the database	Use the concept of the SAP Logical Unit of Work (LUW) integrated into the ABAP runtime environment which enables multiple users to access the central database at the same time.
Using external interfaces	Use Remote Function Call as an interface to other programming environments. Use the Internet Communication Framework (ICF) as an interface to the Internet. Use the built-in transformation capabilities to support XML and JSON.
Supporting UIs in multiple languages	Extract language-specific program components from the source code, which are reloaded when the program executes in accordance with the environment.

Data Modeling

Data modeling provides capabilities for defining data models in the database that can be used for business applications.

Key Feature	Use
ABAP Dictionary	Maintain data types in a persistent repository that are visible in all repository objects. The database tables of the central database, views, and lock objects are managed in ABAP Dictionary.
ABAP Core Data Services (CDS)	Implement SAP's CDS concept for AS ABAP. The concept provides a platform-independent Data Definition Language (DDL) to define CDS entities that implement a semantically rich data model.

Database Access

Database access provides capabilities for accessing database entities.

Key Feature	Use
ABAP SQL	Access data through ABAP SQL statements integrated into the ABAP programming language. The ABAP statements allow access to ABAP-managed database objects in SAP HANA defined in the ABAP Dictionary or by ABAP CDS.

Key Feature	Use
Native SQL	Platform-dependent access to a database. Native SQL includes an embedded static access (EXEC SQL ABAP statement), a class-based, general purpose access in SAP HANA using ABAP Database Connectivity, and ABAP-managed database procedures.

ABAP RESTful Application Programming Model (RAP)

The ABAP RESTful Application Programming Model (RAP) supports the model-driven development of transactional enterprise SAP Fiori applications and Web APIs that are intrinsically optimized for SAP HANA. RAP is based on technologies and frameworks such as Core Data Services (CDS) for defining semantically rich data models, the ABAP language enhanced with the Entity Manipulation Language (EML) for implementing the custom business logic, and a business service model infrastructure for creating business services.

Key Feature	Use
Developing brownfield applications	Develop OData services (V2 or V4) using existing code with the unmanaged implementation type. Draft handling is supported.
Developing greenfield applications	Develop OData services (V2 or V4) from scratch with the managed implementation type based on determinations/validations. Draft handling is supported.
Creating data models for entities (read-only access) and business objects (transactional access) using CDS views	A data model that defines the data structure explicitly, such as relations between datasets, data semantics, and their restrictions.
Defining the behavior of business objects	Define the lifecycle operations on the data, such as create, update and delete, further operations such as actions and transactional properties of the data model.
Implementing business objects	Supply the business logic for the behavior definition in ABAP.
Defining service definitions	Define the scope of a business service by exposing the relevant CDS entities.
Creating service binding	Expose a business service by binding its service definition to a protocol and a scenario. Consume a business service via OData (V2 or V4) or InA protocol or as SQL - Web API.
Developing RAP Business Events	Define and raise RAP Business Events in a business object.

ABAP Programming Model for SAP Fiori

The ABAP Programming Model for SAP Fiori (Application Infrastructure) adopts technical details from the application development and improves the overall development process.

New applications shall be implemented using the ABAP RESTful Application Programming Model (RAP).

Key Feature	Use
Exposing CDS views through OData	Generate OData services based on CDS views without any additional implementation.
Supporting transactional applications on a stateless protocol via draft	Expose CDS-based BOPF business objects through stateless OData services.

Business Object Processing Framework (BOPF)

The Business Object Processing Framework is an ABAP object-oriented framework that provides a set of generic services and functionalities to speed up, standardize, and modularize your development. BOPF manages the entire life cycle of your business objects and covers all aspects of your business application development.

New applications shall be implemented using the ABAP RESTful Application Programming Model (RAP).

Key Feature	Use
Creating and defining business objects	Create new business objects in order to encapsulate business logic and data to be exposed through common interface. Use BOPF tools for developing and testing business objects.
Enhancing business objects	Enhance existing business objects through custom business logic and data.
Integrating as well as combining business objects and non-BOPF applications	Invoke application logic that is not built on BOPF through BOPF objects and vice versa.
Reusing application infrastructure	Reduce development and maintenance effort through reuse of existing BOPF services.

Server Technology

The components of the infrastructure server technology provide a reusable runtime environment for other components that also uses client server technology to process requests.

Key Feature	Use
Application Server ABAP	Use the ABAP runtime environment as a virtual machine for ABAP programs.
Building globalized custom applications	Build globalized custom applications using internationalization technologies and data encodings.
Internationalization	Provides technologies and data encodings for building globalized custom applications.

ABAP Development Tools

The ABAP development tools enable ABAP developers to build their own applications and features using the ABAP programming language and the ABAP Application Server capabilities in an Eclipse-based IDE or on the classic SAP GUI-based ABAP Workbench.

Key Feature	Use
Creating and managing data definitions	Specify ABAP coding such as reports and includes, function modules, classes, and interfaces. Use ABAP Dictionary artifacts such as structures and data elements to create and manage data definitions (metadata).
Rearranging table columns to application-specific needs	Build CDS views on existing database tables using Core Data Services.
Converting between ABAP data and XML formats	Describe transformations between ABAP data and XML formats using transformations (XSLT and ST).
Error handling	Use options such as message classes to inform the user about an error or a status, or to issue a warning
Developing SAP standard UI technology	Use the supported UI Technologies [page 222] to develop SAP solutions
Modularizing and encapsulating development objects	Modularize SAP systems and encapsulate development objects using packages and package interfaces.
Managing the lifecycle of development activities	Modularize, encapsulate, and decouple units in the SAP System using a Package Builder. Transport objects to other SAP systems.
Creating and managing workflow scenarios	Build scenarios to enable flexible workflow creation, and prepare an environment for modeling workflows

ABAP Testing and Quality Tools

The AS ABAP offers a set of specialized test and analysis tools that are useful at different phases of the software lifecycle (development, testing, and production).

Key Feature	Use
Checking code quality	Use static checks of the ABAP Test Cockpit (ATC) and the Code Inspector to analyze customer code for functional, usability, or performance issues. Use the remote ATC checks to verify code quality on older SAP NetWeaver releases with the latest check version. Use static checks of the Code Vulnerability Analyzer (extra license required) to analyze customer code for possible security vulnerabilities.
Unit testing ABAP programs	Use ABAP Unit test classes to develop and execute test classes that ensure the correct behavior of program logic.

ABAP Troubleshooting Tools

The ABAP Troubleshooting Tools enable you to detect functional and performance issues through debugging, tracing, monitoring, and logging applications.

Key Feature	Use
Debugging	Set breakpoints and step through a running ABAP application, analyze and change variable values, and examine the call stack to understand the program integration in the general application context. Debug ABAP-managed database procedures (AMDP) and simple transformations.
Profiling ABAP source code	Create ABAP trace files to analyze the distribution of an application's time consumption with respect to database and ABAP runtime. Also, to significantly improve the overall performance and detect hotspots such as: • SQL accesses • ABAP procedures or statements with poor performance • redundant executions of same program parts
Monitoring SQL performance of an ABAP system	Collect essential SQL performance data along with the ABAP contexts of all SQL statements performed in an ABAP system. Analyze the collected data for time intervals of your choice.
Dynamic logging	Create dynamic logpoints for specific code positions to gain runtime information like the number of executions, variable values, call stack information, or to use them for triggering system functions like the SQL trace. It is not necessary to change source code in order to use dynamic logpoints.

Extensibility

The extensibility configuration framework allows different interface implementations of the application parts to be exchanged during the runtime of an application.

Key Feature	Use
Create custom fields, data source extensions, and custom logic	Create your own fields and enhancement implementations for specific business contexts of extensible applications. Enable the usage of existing fields in predelivered data sources using data source extensions. Custom Fields • You can translate the fields that you created into different languages and enable field usage for UIs, reports, email templates, form templates, business scenarios, and APIs. • You can make field content relevant for free-text search. • You can choose the aggregation behavior of fields. • You can delete already transported fields.. Data Source Extensions • You can create, edit, and delete data source extensions in order to enable the usage of existing fields in predelivered data sources. Custom Logic • You can implement custom logic with ABAP for key users in your enhancement implementation. • You can create and save variants for testing custom logic with predefined parameters. • You can create and save filters to define under which conditions the logic of an enhancement implementation is used. You can publish fields and enhancement implementations and thus generate them in the extensible applications in your test system. You can edit fields and enhancement implementations even after they have already been published. You can delete fields and enhancement implementations.
Manage custom logic after upgrades	Adapt enhancements and modifications after an upgrade or support package implementation due to changes in SAP applications.
Create enhancement options	Create, edit, and delete enhancement options for Business Add-ins (BAI) or source code plug-ins to make custom applications for ABAP extensible.

Key Feature	Use
Customizing analytical queries	Design a meaningful query for analyzing and reporting purposes without knowing query language and other technical details. Perform various operations such as create a new query, copy from the existing query, modify the query, add/remove custom fields, adding dimensions and measures from the data view, defining filters, and preview result set.
Create custom reusable elements	You can create reusable custom libraries for consumption in custom logic extensions or custom objects. You can create translatable custom code lists for reuse across custom business objects.
Create custom CDS views	You can create and maintain custom CDS views based on the virtual data model delivered by SAP. You can define projections and add associations to your custom CDS views. You can maintain parameters, filters and calculated fields. You can define custom CDS views without a scenario or as cubes, dimensions or for usage as external API.
Create traces	You can create traces to track the processing of custom logic in custom objects, custom reusable elements and enhancement implementations.
Create custom business objects	Create new custom business objects, and generate OData services based on custom business objects. You can enable associations between your custom business objects. You can add fields to and delete fields from custom business objects. You can add multiple subnodes to a custom business object, and implement custom logic with ABAP for key users. You can edit and publish a custom business object, and delete the draft of a custom business object. You can delete custom business objects.
Display list of extension items	You can view a list of your extensibility items, and the dependencies between them.
View namespaces	You can view namespaces which were used for extension item development.
Configure software packages	You can configure software packages to be used for extension item development.
Register extension items for transport	You can assign extension items to a transport.

2.14.1.4 Security for ABAP

Business Background

In today's world of collaborative business processes and open system environments, security no longer means just adding a firewall and using passwords to log on. It requires a complete approach that not only applies to your own IT landscape, but also to issues that arise beyond your own borders, in which even simple organizational measures can have a significant impact. The infrastructure of the ABAP platform supports you by delivering comprehensive security features for heterogeneous environments.

Key Features

The following features are available:

Key Feature	Use
Authorization management	Assign authorizations to users with roles. Create roles. Mass maintenance enables authorization management for many users. User information system allows for reporting of roles and their authorizations.
Identity management	Define user validity. Maintain user master data. Set user types. Support for mass maintenance of user data. Delegated user administration. User information system allows for reporting of users. Synchronize users data with other systems.
Central user administration	Push users and authorizations from a central system in a system landscape.
Security policy	Define security policies to control password policies, logon policies, and policies for logon IDs.

Key Feature	Use
User authentication and single sign-on	Configure the authentication methods and policies supported by the system. For the following scenarios, SAP NetWeaver Application Server supports the following authentication and single sign-on (SSO) methods: • SAP GUI logon • Password-based authentication • SSO for SAP Shortcuts • SSO with an external SNC product • Web-based logon • Password-based authentication • SSO with logon tickets • SSO with X.509 client certificates • Kerberos with SPNego Requires a license with SAP Single Sign-On to support • SSO with external SAML 2.0 identity provider • SOAP Web Services • Transport level authentication and SSO with user ID and password, X.509 certificates, and authentication assertion tickets. • Message level authentication using user name token, X.509 certificate token, and SAML token. You can also use WS-specific security and authentication mechanisms, such as XML encryption, XML signatures, Message Aging, and WS SecureConversation • OAuth 2.0 authorization framework • Acts as both resource server and authorization server in the same system for OData requests Can act as client • Log off methods where possible
Digital signatures and encryption	Encrypt data during transport: Support of transport layer security (TLS), secure network communication (SNC), and web service security. Digitally sign and verify signed documents. Manage digital keys.